MILITARY SPECIFICATION

LUBRICANT, DRY THIN FILM, MOLECULAR BONDED

This amendment forms a part of DOD-L-85645A, dated 15 October 1988, and is approved for use by all Departments and Agencies of the Department of Defense

PAGE 2

2.1.2, DELETE title of FED-STD-313 and SUBSTITUTE "Material Safety Data, Transportation, and Disposal Data for Hazardous Material Furnished to Government Activities".

Page 10

APPENDIX: DELETE in its entirety and SUBSTITUTE:

"APPENDIX

INSTRUCTIONS FOR APPLYING LUBRICANT, DRY THIN FILM, MOLECULAR BONDED

10. SCOPE

10.1 Scope. This appendix covers the application of the dry film lubricant when it is applied over various metal surfaces. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

20. APPLICABLE DOCUMENTS

20.1 Government documents. The following documents form a part of this Appendix to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-T-634 - Trichloroethylene, Technical

TT-N-95 - Naphtha, Aliphatic

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Inspection. All parts, prior to being degreased, shall be free of all foreign matter. Assemblies with plastic or other non metallic components shall be degreased only when authorized by the contract or applicable drawing. Parts that are excessively soiled or contaminated shall undergo a preliminary cleaning in a steam bath, or soak in a detergent type cleaner providing such treatment is authorized by the contract or applicable drawing.

30.2 Cleaning. Oils, greases, waxes and other foreign matter shall be cleaned from test specimens and parts, with the excepton of titanium, by vapor degreasing with trichloroethylene conforming to O-T-634, Type II, or solvent wiping with a bleached cotton cheesecloth soaked with aliphatic naphtha conforming to TT-N-95. Titanium and titanium alloys shall only be cleaned by solvent wiping with aliphatic naphtha using bleached cotton cheesecloth. The surface shall be air blasted with a suitable mesh powder that can provide a uniform clean surface. The blast shall not exceed 690 kPa and will be directed in such a manner as to not change the surface dimensional tolerances.

30.3 General instructions for all metals. All metal surfaces shall be treated or coated in accordance with the applicable specification or drawing before applying the lubricant.

30.4 Application. The lubricant is applied through an air blast gun at 828 kPa, 25.4 to 30.5 cm from the part. Only one coat of dry film lubricant can be applied. Material will not adhere to itself for additional buildup. The control thickness ranges from 0.00038 and 0.00051 mm, but should not exceed 0.0025 mm unless otherwise specified in the contract or purchase order and determined in accordance with ASTM D1186.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

30.5 Curing. Curing shall be in accordance with the applicable classification. Recommended cure cycles are as follows:

Type I: No curing required.

Type II: Air Dry: Seven days at ambient temperature (25°C nominal) before using part in its intended application.

Oven Cure: 149°C for 2.0 hrs.

30.6 Post-treatment. No post-treatments are recommended unless approved by the procuring activity."

Custodian:

Navy - AS
Air Force - 11
Army - ME

Review activity:
DLA - GS
Army - SM

User:
Army - AR

Preparing activity:
Navy - AS

(Project No. 9150-1115)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.